Hyperspectral Image Denoising via Tensor Low-Rank Prior and Unsupervised Deep Spatial–Spectral Prior

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***Abstract*— Hyperspectral image (HSI) denoising is a fundamental task in remote sensing image processing, which is helpful for HSI subsequent applications, such as unmixing and classification. Thanks to the powerful representation ability of untrained deep neural networks (DNNs), deep image prior (DIP)-based methods achieve tremendous successes in image processing (e.g., denoising and inpainting). However, DIP-based methods neglect the tensor low-rank prior of the underlying HSI, which will be beneficial to capture the global structure of the underlying HSI. To address this issue, we propose a novel model for HSI denoising, which can simultaneously take respective advantages of the tensor low-rank prior and the deep spatial–spectral prior. The tensor low-rank prior leads to a better global structure, and the deep spatial–spectral prior is complementary to preserve better local details. On the one hand, we adopt low-rank tensor ring (TR) decomposition to characterize the tensor low-rank prior and capture the global structure of the underlying HSI. On the other hand, we employ untrained DNNs to flexibly represent the deep spatial–spectral prior and capture the local details of the underlying HSI. To solve the proposed model, we develop an efficient alternating minimization algorithm. Experimental results on simulated and real data validate the advantages of the proposed model in HSI denoising. Compared with state-of-the-art HSI denoising methods, the proposed method preserves better local details and the global structure of the underlying HSI.**

***Index Terms*— Deep spatial–spectral prior, tensor low-rank prior, tensor ring (TR) decomposition.**

I. Introduction

HYPERSPECTRAL images (HSIs), which are essentially three-dimensional arrays, have been widely used in earth observation, urban planning, and agriculture [1], [2], [3], [4] because they have abundant spatial and spectral information. Nevertheless, owing to sensor instability, atmosphere interference, and photon effects, HSIs often suffer from various types of noise, such as Gaussian noise, salt and pepper noise, stripes, and deadlines [5], [6], [7]. The noise severely reduces the image quality, which would impair the performance of subsequent applications, including classification and change detection [8], [9], [10], [11]. Thus, HSI denoising is a fundamental task in remote sensing image processing. HSI denoising is a classical ill-posed inverse problem. The key to solve this ill-posed inverse problem is to deeply explore and fully exploit the potential priors of HSIs [12], [13], [14].

In the past few decades, low-rank prior [15], [16] has been widely applied for HSI denoising, on the basis of the hypothesis that the underlying HSIs commonly exist in a low-dimensional subspace [17]. Previous studies unfolded HSIs to matrices and extended low-rank matrices recovery methods to HSI denoising [18], [19], [20]. For example, Zhang et al. [21] ordered the patch of the HSI into a matrix lexicographically and proposed a low-rank matrix recovery model for HSI denoising. In view of the piecewise smooth structure of HSIs, He et al. [22] proposed a total variation (TV)-regularized low-rank matrix decomposition model for HSI denoising, which obtains the inspiring performance. Xie et al. [23] utilized a nonconvex low-rank regularizer (i.e., the weighted Schatten p-norm) to gain a better low-rank approximation. However, unfolding an HSI to a matrix unavoidably destroys the intrinsic structure of the HSI. To overcome this drawback, many tensor low-rank-based methods have been proposed, which have achieved great success in HSI processing [24], [25], [26]. For instance, Liu et al. [27] proposed a multilinear algebra model, named parallel factor analysis (PARAFAC), to reconstruct HSIs. Considering the nonlocal self-similarity of the underlying HSI, Xue et al. [28] proposed a nonlocal regularized CP decomposition model for HSI denoising. Based on the Tucker decomposition [29], Renard et al. [30] proposed a multilinear algebra method, which processes spatial and spectral dimensions jointly. Considering the piecewise smooth structure of HSIs, Wang et al. [31] proposed a TV regularized low-rank tensor decomposition model for HSI denoising, in which Tucker decomposition is utilized to capture the global structure, and TV is utilized to capture the piecewise smooth structure. Based on tensor singular value decomposition (t-SVD), Zheng et al. [32] proposed a fibered rank minimization model for HSI denoising. Zeng et al. [33] proposed $L_{1\text{-}2}$ spatial–spectral TV (SSTV) regularized low-rank tensor decomposition model for HSI denoising, which utilizes $L_{1\text{-}2}$ SSTV to capture the piecewise smooth structure and t-SVD to

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characterize the global spatial–spectral correlation. It is worth noting that TR decomposition is a more fundamental decomposition than other decompositions, in which the core tensors can be circularly shifted and treated equally. Chen et al. [34] introduced tensor ring (TR) decomposition into HSI denoising and proposed a nonlocal TR decomposition model, which exploits the nonlocal self-similarity and the global spectral correlation to achieve satisfactory results. Moreover, to more effectively exploit the low-rank characteristic of the underlying HSI, Chen et al. [35] proposed two different nonconvex low-rank approximation models for HSI denoising, which utilize factor group sparsity regularizer to further enhance the low-rank property.

Recently, deep learning-based methods have received increasing attention, thanks to the powerful representation ability of deep neural networks (DNNs), which map the observed HSI to the underlying HSI. Based on whether a large number of paired training data (noisy image and the corresponding clean image) are required, deep learning-based methods can be divided into two groups, i.e., supervised learning-based methods and unsupervised learning-based methods.

Supervised learning-based methods usually train DNNs [36], [37] to recover a noise-free HSI from a noisy observation, which needs a good deal of paired training data as the basis. Chang et al. [38] introduced the deep convolutional neural network (CNN) for HSI mixed noise removal, in which the learned filters are able to extract the spatial information in the local region and the channels of these filters can describe the spectral correlation. In order to fully utilize the spatial–spectral correlation, Dong et al. [39] proposed a 3-D CNN-based model for HSI denoising and employed the separable 3-D convolution to substantially reduce the number of network parameters. To adapt multilevel noise better, Yuan et al. [40] utilized a noise estimation network to obtain the noise level and trained a multiscale residual network for HSI denoising. The aforementioned methods have acquired attractive results in HSI denoising. However, these methods depend on the diversity and volume of the paired training dataset, which is not always available for HSIs.

Unsupervised learning-based methods utilize the deep image prior (DIP) expressed by suitable network architecture to solve various image recovery problems without any paired training data [41], [42]. Employing unsupervised learning-based methods for HSI processing can effectively avoid the insufficient training data problem; thus, unsupervised learning-based methods have received increasing attention and rapidly developed. Ulyancv et al. [43] proposed an unsupervised image recovery framework named DIP. The work in [43] showed that suitable network structures can extract the complex structures of the underlying HSI from the input noisy image itself without any extra training data. Sidorov and Hardeberg [44] extended the DIP to HSI processing (e.g., denoising, inpainting, and super-resolution) by replacing all the 2-D convolution with 3-D convolution. Inspired by the linear mixture model, Miao et al. [45] proposed an unsupervised HSI denoising framework, which is guided by disentangled spatiospectral deep priors. These DIP-based methods have good performances in HSI denoising and preserve the local details of the underlying HSI well. However, they neglect the intrinsic low-rank structure of the underlying HSI, which can be captured by the aforementioned low-rank tensor decompositions. A natural question is whether the tensor low-rank prior, and the unsupervised deep prior can be organically integrated to benefit each other and improve the performance of HSI denoising. This article provides a positive answer.

To adequately exploit the respective strengths of the tensor low-rank prior and the deep spatial–spectral prior, in this article, we propose a novel model named LRDS2 for HSI mixed noise removal. On the one hand, we adopt low-rank TR decomposition to capture the global spatial–spectral correlation of the underlying HSI. On the other hand, we employ the deep spatial–spectral prior expressed by U-Net [46] networks and fully connected networks (FCNs) [47] to capture the local details of the underlying HSI, which are hardly captured by TR decomposition.

We illustrate the impact of the tensor low-rank prior and deep spatial–spectral prior in the proposed LRDS2 on the performance of HSI denoising in Fig. 1. We observe that the recovered result by the proposed LRDS2 contains more geometrical features compared with LRDS2 without the deep spatial–spectral prior. The reason behind this observation is that untrained DNNs have a powerful representation ability, which is helpful to recover the local details of the underlying HSI. We can also observe that the spectral vector of the denoising result by the proposed LRDS2 is closer to the ground truth compared with LRDS2 without the tensor low-rank prior. The underlying reason is that the tensor low-rank prior can effectively characterize the global spatial–spectral correlation, which leads to a significant improvement in preserving the global structure [48]. These observations empirically verify that the tensor low-rank prior and deep spatial–spectral prior can be organically integrated to benefit each other.

The main contributions of this article are summarized as follows.

1) We propose a novel model for HSI mixed noise removal (named LRDS2), which can simultaneously take respective advantages of the tensor low-rank prior expressed by TR decomposition and the deep spatial–spectral prior expressed by untrained DNNs.
2) We develop an efficient alternating minimization algorithm to solve the proposed LRDS2. Comprehensive experiments on simulated and real data demonstrate that the proposed LRDS2 outperforms the state-of-the-art methods quantitatively and visually.

The rest of this article is organized as follows. In Section II, we introduce the notations and some definitions used in this article. In Section III, we give the details of the proposed LRDS2 and the corresponding solving algorithm. In Section IV, we report and analyze the experimental results on simulated and real data. In Section V, we present some discussions. In Section VI, we conclude this article.

II. Notations and Preliminaries

In this article, scalars, vectors, matrices, and tensors are denoted as x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$, respectively. The ith, (i, j)th, and

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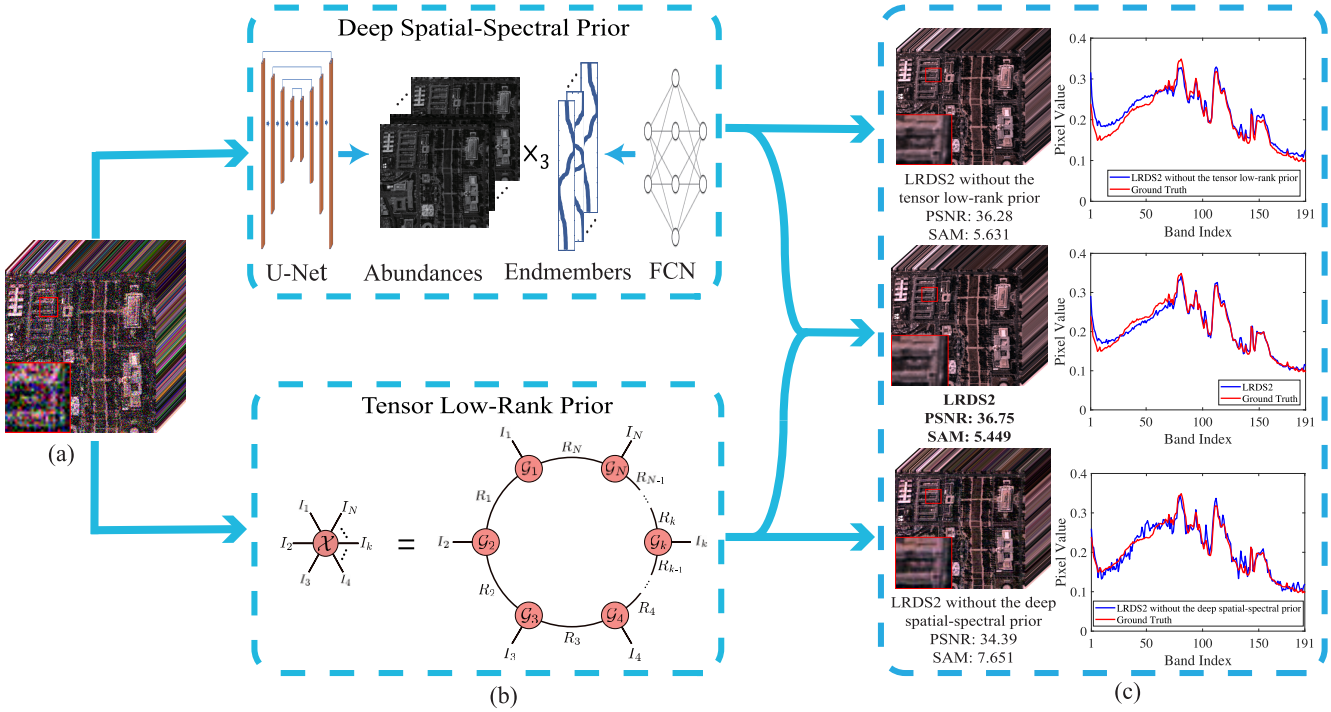


Fig. 1. Flowchart of the proposed LRDS2. From (left) to (right): (a) observed noisy HSI, (b) deep spatial–spectral prior expressed by untrained U-Net networks and FCNs and the tensor low-rank prior expressed by TR decomposition in LRDS2, and (c) denoising results by LRDS2 without the tensor low-rank prior, LRDS2, and LRDS2 without the deep spatial–spectral prior. Here, the displayed denoising images are false-color images composed of bands (25, 7, 11), and the spectral vectors of the denoising results are at the spatial location (192, 2).

(i, j, k)th elements of $\mathbf{x} \in \mathbb{R}^I$, $\mathbf{X} \in \mathbb{R}^{I\times J}$, and $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ are denoted as $\mathbf{x}_i$, $\mathbf{X}_{i,j}$, and $\mathcal{X}_{i,j,k}$, respectively. $\mathbf{X}(j) \in \mathbb{R}^{I\times K}$ denotes the jth lateral slice of $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ whose (i, k)th entry satisfies $\mathbf{X}(j)_{i,k} = \mathcal{X}_{i,j,k}$. $\|\mathcal{X}\|_F = (\sum_{i,j,k} \mathcal{X}^2_{i,j,k})^{1/2}$ denotes the Frobenius norm of $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$. The first mode-$n$ unfolding [31] is denoted as $\mathbf{X}_{(n)} \in \mathbb{R}^{I_n\times I_1\cdots I_{n-1}I_{n+1}\cdots I_N}$, which maps the $(i_1, i_2, \ldots, i_N)$th entry of $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ to the $(i_n, i_1 + \sum_{s=2}^{n-1}((i_s - 1)\prod_{m=1}^{s-1} I_m) + \sum_{s=n+1}^{N}((i_s - 1)\prod_{m=n+1}^{s-1} I_m)\prod_{l=1}^{n-1} I_l)$th entry of $\mathbf{X}_{(n)}$. The second mode-n unfolding [49] is denoted as $\mathbf{X}_{<n>} \in \mathbb{R}^{I_n\times I_{n+1}\cdots I_N I_1\cdots I_{n-1}}$, which maps the $(i_1, i_2, \ldots, i_N)$th entry of $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ to the $(i_n, i_{n+1} + \sum_{s=n+1}^{N}((i_s - 1)\prod_{m=n+1}^{s-1} I_m) + \sum_{s=1}^{n-1}((i_s - 1)\prod_{m=1}^{s-1} I_m)\prod_{l=n+1}^{N} I_l)$th entry of $\mathbf{X}_{<n>}$.

Definition 1 (TR Decomposition [49]): TR decomposition that is denoted as $\mathcal{X} = \Phi(\mathcal{G})$ represents a tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ by circular multilinear products over a sequence of third-order core tensors $\mathcal{G} = \{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \ldots, \mathcal{G}^{(N)}\}$, where $\mathcal{G}^{(n)} \in \mathbb{R}^{r_{n-1}\times I_n\times r_n}$, $n = 1, 2, \ldots, N$, and $r_0 = r_N$. The vector $[r_1, r_2, \ldots, r_N]$ is called TR rank. For TR decomposition, the $(i_1, i_2, \ldots, i_N)$th element of $\mathcal{X}$ satisfies

$$\mathcal{X}_{i_1,i_2,\ldots,i_N} = \text{Trace}\left(\mathbf{G}^{(1)}(i_1)\mathbf{G}^{(2)}(i_2)\cdots\mathbf{G}^{(N)}(i_N)\right) \tag{1}$$

where Trace$(\cdot)$ is the matrix trace operation.

Definition 2 (Multilinear Product [49]): The multilinear product of $\mathcal{G}^{(n)} \in \mathbb{R}^{r_{n-1}\times I_n\times r_n}$ and $\mathcal{G}^{(n+1)} \in \mathbb{R}^{r_n\times I_{n+1}\times r_{n+1}}$ is denoted $\mathcal{G}^{(n,n+1)} \in \mathbb{R}^{r_{n-1}\times I_n I_{n+1}\times r_{n+1}}$, and its $i_k + I_{n+1}(j_k - 1)$th lateral slice satisfies

$$\mathbf{G}^{(n,n+1)}(i_k + I_n(j_k - 1)) = \mathbf{G}^{(n)}(i_k)\mathbf{G}^{(n+1)}(j_k) \tag{2}$$

for $i_k = 1, 2, \ldots, I_n$ and $j_k = 1, 2, \ldots, I_{n+1}$.

Definition 3 (Matrix-Vector Outer Product [49]): The outer product of matrix $\mathbf{X} \in \mathbb{R}^{I_1\times I_2}$ and vector $\mathbf{y} \in \mathbb{R}^{I_3}$ is $\mathcal{Z} = \mathbf{X}\circ\mathbf{y} \in \mathbb{R}^{I_1\times I_2\times I_3}$, and its (i, j, k)th element satisfies $\mathcal{Z}_{i,j,k} = \mathbf{X}_{i,j}\mathbf{y}_k$.

III. Proposed Model

A. Proposed Method

The HSI degradation model can be formulated as the following additive model:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{3}$$

where $\mathcal{Y} \in \mathbb{R}^{I_1\times I_2\times I_3}$, $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$, $\mathcal{S} \in \mathbb{R}^{I_1\times I_2\times I_3}$, and $\mathcal{N} \in \mathbb{R}^{I_1\times I_2\times I_3}$ denote the noisy HSI, the clean HSI, the sparse noise (salt and pepper noise, deadlines, and stripes), and the Gaussian noise, respectively. Normally, I_1, I_2, and I_3 denote the spatial height, the spatial width, and the spectral number, respectively.

To adequately exploit the respective strengths of the tensor low-rank prior and the deep spatial–spectral prior, we propose a novel model (i.e., LRDS2) for HSI mixed noise removal as follows:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{G},\mathcal{S},\{\theta_m,\xi_m\}_{m=1}^M} \frac{1}{2}||\mathcal{Y} - \mathcal{X} - \mathcal{S}||_F^2 + \mu\|\mathcal{S}\|_1 \\ &\text{s.t.} \qquad \mathcal{X} = \Phi(\mathcal{G}), \quad \mathcal{X} = \sum_{m=1}^{M} \mathbf{A}_{\theta_m}(\mathbf{z}_m)\circ\mathbf{c}_{\xi_m}(\mathbf{w}_m) \end{aligned} \tag{4}$$

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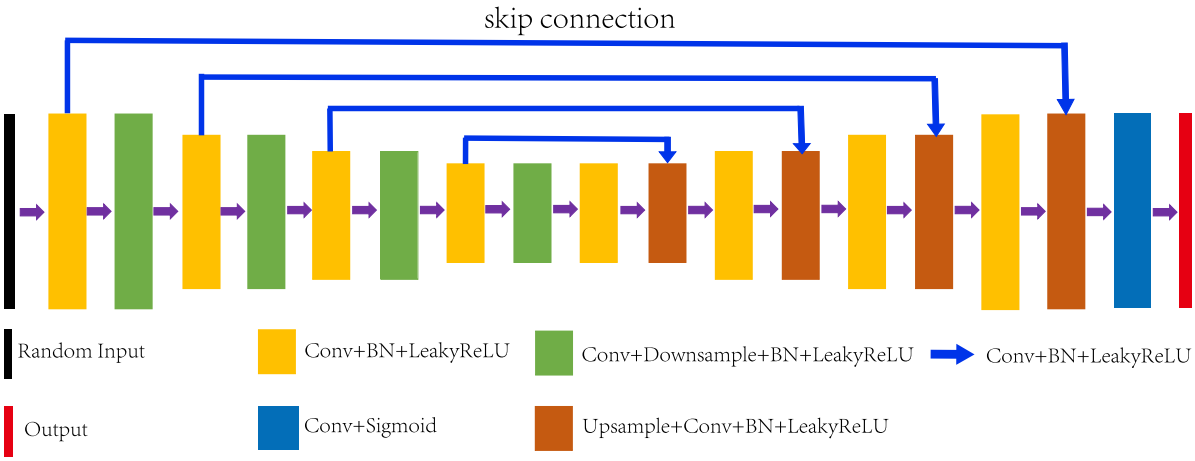


Fig. 2. Architecture of U-Net networks used in our method.

TABLE I
PARAMETERS OF NEURAL NETWORKS USED IN OUR METHOD

Parameters	Values
n_d	$[16, 32, 64, 128, 128, 128]$
n_u	$[128, 128, 128, 64, 32, 16]$
n_s	$[4, 4, 4, 4, 4, 4]$
n_f	$[128, 256, 256, 128]$
k_d	$[7, 7, 5, 5, 3, 3]$
k_u	$[3, 3, 5, 5, 7, 7]$
k_s	$[1, 1, 1, 1, 1, 1]$

where μ is the regularizer parameter, and $\mathbf{A}_{\theta_m}(\mathbf{z}_m) \in \mathbb{R}^{I_1 \times I_2}$ and $\mathbf{c}_{\xi_m}(\mathbf{w}_m) \in \mathbb{R}^{I_3}$ are the mth abundance map and corresponding signature of endmember, which are generated by the U-Net network and FCN, respectively. $\mathbf{z}_m$ and $\mathbf{w}_m$ represent random input of the corresponding U-Net network and FCN. The architecture of U-Net networks used in the proposed method is displayed in Fig. 2. Moreover, more details of neural networks used in our method are reported in Table I. In Table I, n_u, n_d, and n_s denote the number of channels of the convolution layer for the upsampling, downsampling, and skip connections, respectively. k_u, k_d, and k_s denote the corresponding sizes of the convolutional kernels of the convolution layer for upsampling, downsampling, and skip connections, respectively. n_f denotes the number of output channels of the fully connected layer. $\{\theta_m, \xi_m\}_{m=1}^M$ represent the weight of DNNs, and M is the number of endmembers. $\mathcal{X} = \Phi(\mathcal{G})$ denotes the TR decomposition of $\mathcal{X}$ with TR factors $\mathcal{G} = \{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \mathcal{G}^{(3)}\}$. The proposed method enjoys the respective advantages of the tensor low-rank prior and the deep spatial–spectral prior. Specifically, we adopt low-rank TR decomposition to characterize the tensor low-rank prior and capture the global structure of the underlying HSI. Meanwhile, we employ the U-Net networks and FCNs to characterize the deep spatial–spectral prior and capture the local details of the underlying HSI.

B. Alternating Minimization-Based Solving Algorithm

In order to solve the proposed LRDS2, we resort to the half-quadratic splitting (HQS) [50], [51], [52] technique and transform the constrained problem (4) into following unconstrained problem:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{S},\{\theta_m,\xi_m\}_{m=1}^M,\mathcal{G}} \ & \frac{1}{2}||\mathcal{Y}-\mathcal{X}-\mathcal{S}||_F^2+\mu\|\mathcal{S}\|_1 \\ & +\frac{\beta_1}{2}\|\mathcal{X}-\Phi(\mathcal{G})\|_F^2 \\ & +\frac{\beta_2}{2}\|\mathcal{X}-\sum_{m=1}^{M}\mathbf{A}_{\theta_m}(\mathbf{z}_m)\circ\mathbf{c}_{\xi_m}(\mathbf{w}_m)\|_F^2 \end{aligned} \tag{5}$$

where β_1 and β_2 are penalty parameters. Within the framework of alternating minimization algorithm [53], the problem (5) can be solved by iteratively updating each variable as follows.

1) Updating $\mathcal{G}^{(1)}$: Fixing other variables, the $\mathcal{G}^{(1)} \in \mathbb{R}^{r_1 \times I_1 \times r_2}$ subproblem is

$$\min_{\mathcal{G}^{(1)}}\left|\left|\mathbf{X}^t_{<1>}-\mathbf{G}^{(1)}_{<2>}\left(\mathbf{G}^{(2,3),t}_{<2>}\right)^T\right|\right|_F^2 \tag{6}$$

where $\mathbf{G}^{(1)}_{<2>}$ is the second mode-2 unfolding of $\mathcal{G}^{(1)}$ and $\mathbf{G}^{(2,3)}_{<2>}$ is the second mode-2 unfolding of the result of multilinear product of $\mathcal{G}^{(2)}$ and $\mathcal{G}^{(3)}$. $\mathbf{X}_{<1>}$ is the second mode-1 unfolding of $\mathcal{X}$. The problem (6) can be exactly solved as follows:

$$\mathcal{G}^{(1),t+1}=\text{fold}_2\left(\mathbf{X}^{\text{t}}_{<1>}\mathbf{G}^{(2,3),\text{t}}_{<2>}\times\left(\left(\mathbf{G}^{(2,3),\text{t}}_{<2>}\right)^{\text{T}}\left(\mathbf{G}^{(2,3),\text{t}}_{<2>}\right)\right)^{-1}\right) \tag{7}$$

where $\text{fold}_2(\cdot)$ is the inverse operator of the second mode-2 unfolding. The computational complexity of updating $\mathcal{G}^{(1)}$ at each iteration is $\mathcal{O}(r_1 r_3 I_1 I_2 I_3 + (r_1 r_3)^2 I_2 I_3)$, where $[r_1, r_2, r_3]$ is the TR rank of $\mathcal{X}$.

2) Updating $\mathcal{G}^{(2)}$: Fixing other variables, the $\mathcal{G}^{(2)} \in \mathbb{R}^{r_2 \times I_2 \times r_3}$ subproblem is

$$\min_{\mathcal{G}^{(2)}}\left|\left|\mathbf{X}^t_{<2>}-\mathbf{G}^{(2)}_{<2>}\left(\mathbf{G}^{(3,1),t}_{<2>}\right)^T\right|\right|_F^2 \tag{8}$$

where $\mathbf{G}^{(2)}_{<2>}$ is the second mode-2 unfolding of $\mathcal{G}^{(2)}$ and $\mathbf{G}^{(3,1)}_{<2>}$ is the second mode-2 unfolding of the result of multilinear

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product of $\mathcal{G}^{(3)}$ and $\mathcal{G}^{(1)}$. $\mathbf{X}_{<2>}$ is the second mode-2 unfolding of $\mathcal{X}$. The problem (8) can be exactly solved as follows:

$$\mathcal{G}^{(2),t+1} = \text{fold}_2\left(\mathbf{X}^{\text{t}}_{<2>}\mathbf{G}^{(3,1),\text{t}}_{<2>} \times \left(\left(\mathbf{G}^{(3,1),\text{t}}_{<2>}\right)^{\text{T}}\left(\mathbf{G}^{(3,1),\text{t}}_{<2>}\right)\right)^{-1}\right). \tag{9}$$

The computational complexity of updating $\mathcal{G}^{(2)}$ at each iteration is $\mathcal{O}(r_1r_2I_1I_2I_3 + (r_1r_2)^2I_1I_3)$.

3) Updating $\mathcal{G}^{(3)}$: Fixing other variables, the $\mathcal{G}^{(3)} \in \mathbb{R}^{r_3\times I_3\times r_1}$ subproblem is

$$\min_{\mathcal{G}^{(3)}}\left|\left|\mathbf{X}^t_{<3>} - \mathbf{G}^{(3)}_{<2>}\left(\mathbf{G}^{(1,2),t}_{<2>}\right)^T\right|\right|^2_F \tag{10}$$

where $\mathbf{G}^{(3)}_{<2>}$ is the second mode-2 unfolding of $\mathcal{G}^{(3)}$ and $\mathbf{G}^{(1,2)}_{<2>}$ is the second mode-2 unfolding of the result of multilinear product of $\mathcal{G}^{(1)}$ and $\mathcal{G}^{(2)}$. $\mathbf{X}_{<3>}$ is the second mode-3 unfolding of $\mathcal{X}$. The problem (10) can be exactly solved as follows:

$$\mathcal{G}^{(3),t+1} = \text{fold}_2\left(\mathbf{X}^{\text{t}}_{<3>}\mathbf{G}^{(1,2),\text{t}}_{<2>} \times \left(\left(\mathbf{G}^{(1,2),\text{t}}_{<2>}\right)^{\text{T}}\left(\mathbf{G}^{(1,2),\text{t}}_{<2>}\right)\right)^{-1}\right). \tag{11}$$

The computational complexity of updating $\mathcal{G}^{(3)}$ at each iteration is $\mathcal{O}(r_2r_3I_1I_2I_3 + (r_2r_3)^2I_1I_2)$.

4) Updating $\mathcal{S}$: Fixing other variables, the $\mathcal{S}$ subproblem is

$$\min_{\mathcal{S}}\frac{1}{2}\left|\left|\mathcal{Y} - \mathcal{X}^t - \mathcal{S}\right|\right|^2_F + \mu\|\mathcal{S}\|_1. \tag{12}$$

The closed-form solution of the problem (12) can be obtained by the soft-thresholding operator, that is,

$$\mathcal{S}^{t+1} = \textbf{shrink}\left(\mathcal{Y} - \mathcal{X}^t, \mu\right) \tag{13}$$

where $[\textbf{shrink}(\mathcal{X}, \mu)]_{i,j,k} = \textbf{sign}(\mathcal{X}_{i,j,k}) \cdot \textbf{max}(|\mathcal{X}_{i,j,k}| - \mu, 0)$ and $\textbf{sign}(\cdot)$ is defined as follows:

$$\textbf{sign}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0. \end{cases} \tag{14}$$

The computational complexity of updating $\mathcal{S}$ at each iteration is $\mathcal{O}(I_1I_2I_3)$.

5) Updating $\{\theta_m, \xi_m\}_{m=1}^M$: Fixing other variables, the $\{\theta_m, \xi_m\}_{m=1}^M$ subproblem is

$$\min_{\{\theta_m,\xi_m\}_{m=1}^R}\mathcal{L}(\theta_m, \xi_m) \triangleq \left|\left|\mathcal{X}^t - \sum_{m=1}^R \mathbf{A}_{\theta_m}(z_m) \circ \mathbf{c}_{\xi_m}(\mathbf{w_m})\right|\right|^2_F. \tag{15}$$

The $\{\theta_m, \xi_m\}_{m=1}^M$ subproblem can be solved by many gradient descent methods [54], [55]. In particular, we choose the Adam optimization algorithm [56] to solve the problem (15).

Under the Adam framework, $\{\theta_m, \xi_m\}_{m=1}^M$ are updated as

$$\theta_m^{t+1} = \theta_m^t - \alpha\frac{u^t_{1,m}}{\sqrt{v^t_{1,m}} + \epsilon} \quad \text{and} \quad \xi_m^{t+1} = \xi_m^t - \alpha\frac{u^t_{2,m}}{\sqrt{v^t_{2,m}} + \epsilon}$$

where

$$u^t_{1,m} = \frac{\beta_1 u^{t-1}_{1,m} + (1-\beta_1)\frac{\partial\mathcal{L}}{\partial\theta^t_m}}{1-\beta_1}$$

$$u^t_{2,m} = \frac{\beta_1 u^{t-1}_{2,m} + (1-\beta_1)\frac{\partial\mathcal{L}}{\partial\xi^t_m}}{1-\beta_1}$$

$$v^t_{1,m} = \frac{\beta_2 v^{t-1}_{1,m} + (1-\beta_2)\left(\frac{\partial\mathcal{L}}{\partial\theta^t_m}\right)^2}{1-\beta_2} \quad \text{and}$$

$$v^t_{2,m} = \frac{\beta_2 v^{t-1}_{2,m} + (1-\beta_2)\left(\frac{\partial\mathcal{L}}{\partial\xi^t_m}\right)^2}{1-\beta_2} \quad (m = 1, 2, \ldots, R)$$

where $\alpha = 0.01$ is the learning rate and the parameters ϵ, β_1, and β_2 are set as $\epsilon = 10^{-8}$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$ by following [56]. The computational complexity of updating $\{\theta_m, \xi_m\}_{m=1}^M$ at each iteration for forward propagation is $\mathcal{O}(\Sigma_{i=1}^6(I_1I_2n_u[i-1]n_u[i]k_s[i]^2 + I_1I_2n_d[i-1]n_d[i]k_d[i]^2 + I_1I_2n_s[i]n_u[i]k_s[i]^2)KM)$, where K is the iteration of the Adam algorithm.

6) Updating $\mathcal{X}$: We update $\mathcal{X}$ by solving the following least-squares problem:

$$\min_{\mathcal{X}}\frac{1}{2}\left|\left|\mathcal{Y} - \mathcal{X} - \mathcal{S}^{t+1}\right|\right|^2_F + \frac{\beta_1}{2}\left|\left|\mathcal{X} - \Phi\left(\mathcal{G}^{t+1}\right)\right|\right|^2_F + \frac{\beta_2}{2}\left|\left|\mathcal{X} - \sum_{m=1}^M \mathbf{A}_{\theta_m^{t+1}}(\mathbf{z}_m) \circ \mathbf{c}_{\xi_m^{t+1}}(\mathbf{w}_m)\right|\right|^2_F \tag{16}$$

where $\mathcal{G}^{t+1} = \{\mathcal{G}^{(1),t+1}, \mathcal{G}^{(2),t+1}, \mathcal{G}^{(3),t+1}\}$. The exact solution of $\mathcal{X}$ is

$$\mathcal{X}^{t+1} = \frac{\mathcal{Y} - \mathcal{S}^{t+1} + \beta_1\Phi\left(\mathcal{G}^{t+1}\right) + \beta_2\mathcal{V}}{1 + \beta_1 + \beta_2} \tag{17}$$

where $\mathcal{V} = \sum_{m=1}^M \mathbf{A}_{\theta_m^{t+1}}(\mathbf{z}_m) \circ \mathbf{c}_{\xi_m^{t+1}}(\mathbf{w}_m)$. The computational complexity of updating $\mathcal{X}$ is $\mathcal{O}(r_1r_3I_1I_2I_3)$.

In summary, the computational complexity of the alternating minimization algorithm at each iteration is $\mathcal{O}((r_1r_2 + r_1r_3 + r_2r_3)I_1I_2I_3 + r_1^2r_2^2I_1I_3 + r_1^2r_3^2I_2I_3 + r_2^2r_3^2I_1I_2 + \Sigma_{i=1}^6(I_1I_2n_u[i-1]n_u[i]k_s[i]^2 + I_1I_2n_d[i-1]n_d[i]k_d[i]^2 + I_1I_2n_s[i]n_u[i]k_s[i]^2)KM)$.

The alternating minimization algorithm for solving the proposed LRDS2 is summarized in Algorithm 1.

IV. Experiments

To demonstrate the effectiveness of the proposed LRDS2 for HSI denoising, we conduct a series of experiments on simulated and real data. To comprehensively demonstrate the superiority of the proposed method, we choose five relative methods as the competing methods, including three unsupervised learning-based methods (DIP2D [43], DIP3D [44], and DS2DP [45]) and two tensor low-rank-based methods (LRTDTV [31] and LRTFL0 [57]). All the parameters involved in these competing methods are adjusted by following the authors' suggestions and fine-tuning to acquire the optimal performance. We employ three quantitative measures to evaluate the performance of different methods, which are the peak signal-to-noise ratio (PSNR) [58], the structural similarity (SSIM) [59], and the spectral angle mapper (SAM) [57].

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Algorithm 1 Alternating Minimization Algorithm

Input: The observed HSI $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the parameters β_1, β_2, μ, TR rank $[r_1, r_2, r_3]$, and the number of endmembers M.

Initialize $\mathcal{G}^{(n),0}$ $(n = 1, 2, 3)$ randomly, $\mathcal{X}^0 = \mathcal{Y}$, $t = 0$, and $t_{max} = 1000$;
while $t \leq t_{max}$ **do**
  Updating $\mathcal{G}^{(1),t+1}$ via (7);
  Updating $\mathcal{G}^{(2),t+1}$ via (9);
  Updating $\mathcal{G}^{(3),t+1}$ via (11);
  Updating $\mathcal{S}^{t+1}$ via (13);
  Updating $\{\theta^{t+1}, \xi^{t+1}\}_{m=1}^{M}$ via (15);
  Updating $\mathcal{X}^{t+1}$ via (17);
  Let $t = t + 1$;
end while

Output: The recovered HSI $\mathcal{X}$.

Normally, the larger PSNR and SSIM values, and the smaller SAM values reveal better denoising performance. Before the beginning of the experiment, all the gray values of simulated and real data are rescaled to [0, 1] band-by-band.

In this work, DIP2D, DIP3D, DS2DP, and the proposed method are implemented by the PyTorch framework on the computer with two Intel Xeon Silver 4210R CPUs, 128-GB RAM, and two Nvidia RTX 3090 GPUs. The experiments of LRTDTV and LRTFL0 are executed in MATLAB (2021b) on a computer with an AMD R5-5600X CPU and 32-GB RAM.

A. Simulated Data

To evaluate the denoising performance of the proposed LRDS2, we conduct a series of simulated experiments on two HSIs. One is the Pavia Centre[1] (Pavia) whose original size is $610 \times 340 \times 115$. We remove some obviously corrupted spectral channels and crop a subregion of Pavia with the size $192 \times 192 \times 80$. The other is the Washington DC Mall[2] (WDC) whose original size is $1028 \times 307 \times 191$. We crop a subregion of WDC with size $256 \times 256 \times 191$ in our experiments.

In order to comprehensively simulate the real HSI noise scenarios, we consider the following four kinds of simulated noise.

1) Case 1 (Gaussian Noise): Pixels in all bands are corrupted by zero-mean Gaussian noise, and the standard deviation of Gaussian noise is set to be 0.1.

2) Case 2 (Gaussian Noise + Salt and Pepper Noise): The added Gaussian noise is the same as Case 1. In addition, the salt and pepper noise, which often appears in HSIs, is also considered. The salt and pepper noise is added to all bands, whose percentage is, respectively, set to be 10%.

3) Case 3 (Gaussian Noise + Salt and Pepper Noise + Deadlines): All bands are corrupted by Gaussian noise and salt and pepper noise as Case 2. In addition, 30% bands that are randomly selected are corrupted by deadlines. For each band corrupted by deadlines, the number of deadlines is randomly selected from 10 to 15, and the spatial width of each deadline is randomly set as 1–3 pixels.

4) Case 4 (Gaussian Noise + Salt and Pepper Noise + Stripes): All bands are corrupted by Gaussian noise and salt and pepper noise as Case 2. Besides, 30% bands that are randomly selected are damaged by stripes. For each band corrupted by stripes, the number of the stripes is randomly selected from 10 to 15, the spatial width of each stripe is randomly selected from 1 to 3 pixels, and the value of each stripe is randomly selected from 0.6 to 0.8.

In LRDS2, there are seven parameters that need to be manually fine-tuned, namely, β_1, β_2, μ, $[r_1, r_2, r_3]$, and M. In our experiments, we set $r_1 = r_2 = r_3 = r$. The parameters β_1 and β_2 are set to be 1 and 100 in all experiments. The parameter μ is selected from $\{10, 1, 0.1, 0.07, 0.05, 0.03\}$ to achieve the best PSNR value in all experiments. For Pavia, the TR rank and M are set to be $[r, r, r] = [7, 7, 7]$ and $M = 4$. For WDC, we set $[r, r, r] = [10, 10, 10]$ and $M = 15$.

a) Quantitative comparison: Table II shows the quantitative quality indices obtained by different methods on Pavia and WDC for Cases 1–4. From Table II, we observe that the proposed LRDS2 acquires better performance than the state-of-the-art competing methods in terms of PSNR, SSIM, and SAM values, which can demonstrate the superiority of the proposed method. Concretely, LRDS2 obtains the increment of around 1.2 dB in the PSNR value in comparison to the second-best method (i.e., DS2DP) on WDC for Case 4. Furthermore, LRDS2 has improvements by around 1.8 dB in terms of PSNR compared with LRTDTV and LRTFL0. The improvements in PSNR value are beneficial from the combination of tensor low-rank prior and deep spatial–spectral prior, which are complementary to each other.

The PSNR values of all bands on Pavia and WDC for Cases 1–4 are shown in Fig. 3. As observed, LRDS2 gains the highest PSNR values on almost all bands in all cases. In particular, we observe that LRDS2 obtains the highest PSNR values in all bands on Pavia for Case 2. LRTFL0 and LRTDTV obtain much lower PSNR values than DS2DP in some bands on Pavia for Case 2. DIP2D and DIP3D obtain much lower PSNR values than LRTDTV and LRTFL0 in all bands on Pavia for Case 2. The satisfactory performance of LRDS2 demonstrates that organically combining the tensor low-rank prior and the deep spatial–spectral prior contributes to improving the performance in HSI mixed noise removal.

b) Visual comparison: For visual comparison, Figs. 4 and 5 show the false-color images of the denoising results by different methods on Pavia and WDC for Cases 1–4. From Figs. 4 and 5, we observe that the false-color images obtained by LRDS2 are closer to the ground truth. Some local details of the results by DIP2D and DIP3D on WDC for Case 3 have some oversmooth and blurry areas, especially in the enlarged region. The false-color image of the denoising result by LRTDTV on WDC for Case 3 shows unnatural color. The false-color image of the result by LRTFL0 on WDC for Case 3 still contains some deadlines. DS2DP achieves better performance than LRTFL0, which removes most deadlines but still loses some local details. The false-color image of the result by LRDS2 shows better local details than

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes.

[2] http://lesun.weebly.com/hyperspectral-data-set.html

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TABLE II

QUANTITATIVE COMPARISON OF ALL THE COMPETING METHODS ON THE WDC AND PAVIA. THE **BEST** AND SECOND-BEST VALUES ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY

Dataset	Case	Indexes	Observed	DIP2D	DIP3D	LRTDTV	LRTFL0	DS2DP	LRDS2
Pavia	Case 1	PSNR	19.998	32.895	31.211	33.883	34.093	35.043	**35.843**
		SSIM	0.422	0.925	0.884	0.932	0.935	0.951	**0.957**
		SAM	25.566	3.269	3.899	4.894	3.728	3.372	**3.152**
	Case 2	PSNR	13.659	28.369	26.804	33.622	33.414	34.435	**35.094**
		SSIM	0.187	0.888	0.801	0.918	0.926	0.944	**0.952**
		SAM	38.567	4.059	5.552	5.264	3.920	4.181	**3.089**
	Case 3	PSNR	13.611	28.272	26.086	31.696	31.744	33.743	**34.015**
		SSIM	0.184	0.877	0.754	0.909	0.906	0.941	**0.942**
		SAM	39.260	5.325	8.196	8.747	6.005	5.213	**4.873**
	Case 4	PSNR	13.327	26.736	25.128	33.097	32.928	33.939	**34.505**
		SSIM	0.177	0.863	0.757	0.917	0.919	0.943	**0.946**
		SAM	38.669	6.140	7.388	5.538	4.266	4.089	**3.888**
WDC	Case 1	PSNR	20.000	29.961	28.465	35.607	35.290	36.282	**36.753**
		SSIM	0.431	0.864	0.792	0.956	0.954	0.964	**0.966**
		SAM	32.8452	7.312	8.799	5.704	6.315	5.631	**5.449**
	Case 2	PSNR	13.224	26.013	24.983	34.953	34.777	35.584	**36.034**
		SSIM	0.202	0.798	0.715	0.951	0.949	0.955	**0.965**
		SAM	32.845	9.696	10.938	6.645	6.220	5.827	**5.762**
	Case 3	PSNR	13.221	26.285	25.043	33.918	33.315	34.660	**35.430**
		SSIM	0.200	0.803	0.697	0.940	0.935	0.951	**0.962**
		SAM	46.714	9.693	11.232	8.736	8.312	6.373	**5.963**
	Case 4	PSNR	12.929	25.129	23.485	34.067	33.739	34.642	**35.803**
		SSIM	0.195	0.779	0.648	0.948	0.938	0.950	**0.962**
		SAM	46.574	11.668	14.955	8.242	9.904	6.406	**5.875**

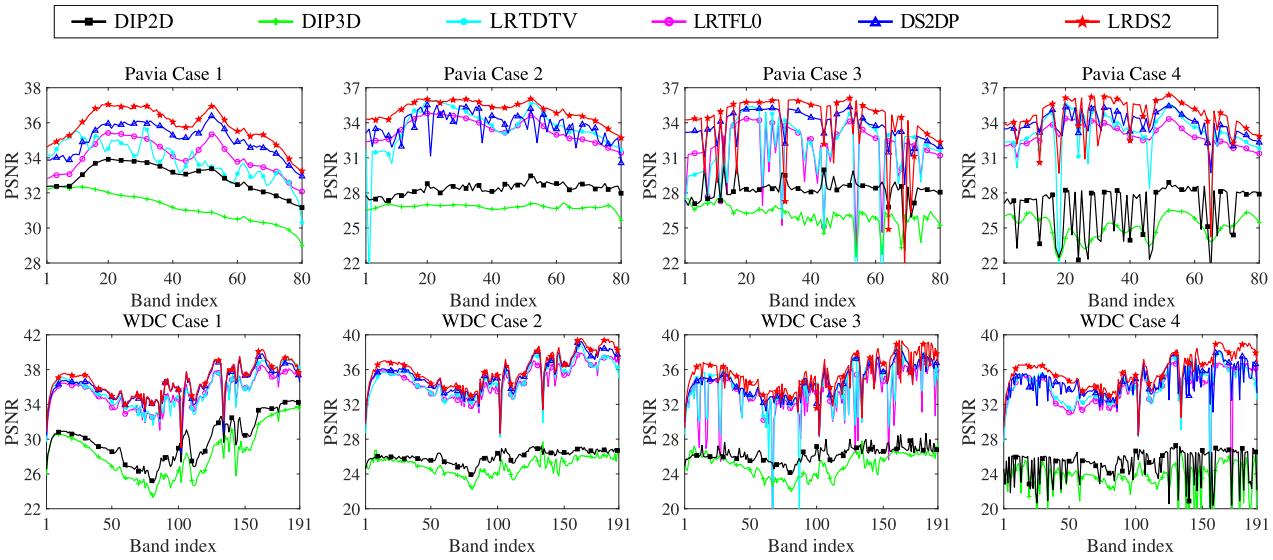


Fig. 3. PSNR values of all bands of the denoising results by different methods on Pavia and WDC for Cases 1–4.

the result obtained by DS2DP, especially in the enlarged region. Moreover, Fig. 6 shows the residual images of the denoising results by different methods on Pavia and WDC for Case 1. From Fig. 6, we can observe that the residual images of the result by LRDS2 are closer to the ground truth, which empirically verifies the superior of the proposed method. These observations reflect that LRDS2 enjoys the advantages of the DNNs, which are used to express the deep spatial–spectral prior and preserve local details well.

For a more comprehensive comparison, Fig. 7 shows the spectral vectors of the denoising results by different methods. We observe that the spectral vectors of the results by the proposed method are closer to the ground truth than tensor low-rank-based methods and unsupervised learning-based methods, which reveals better spectrum fidelity of the proposed method. This is because the tensor low-rank prior and the deep spatial–spectral prior are organically combined in our method, which contributes to improving the denoising performance.

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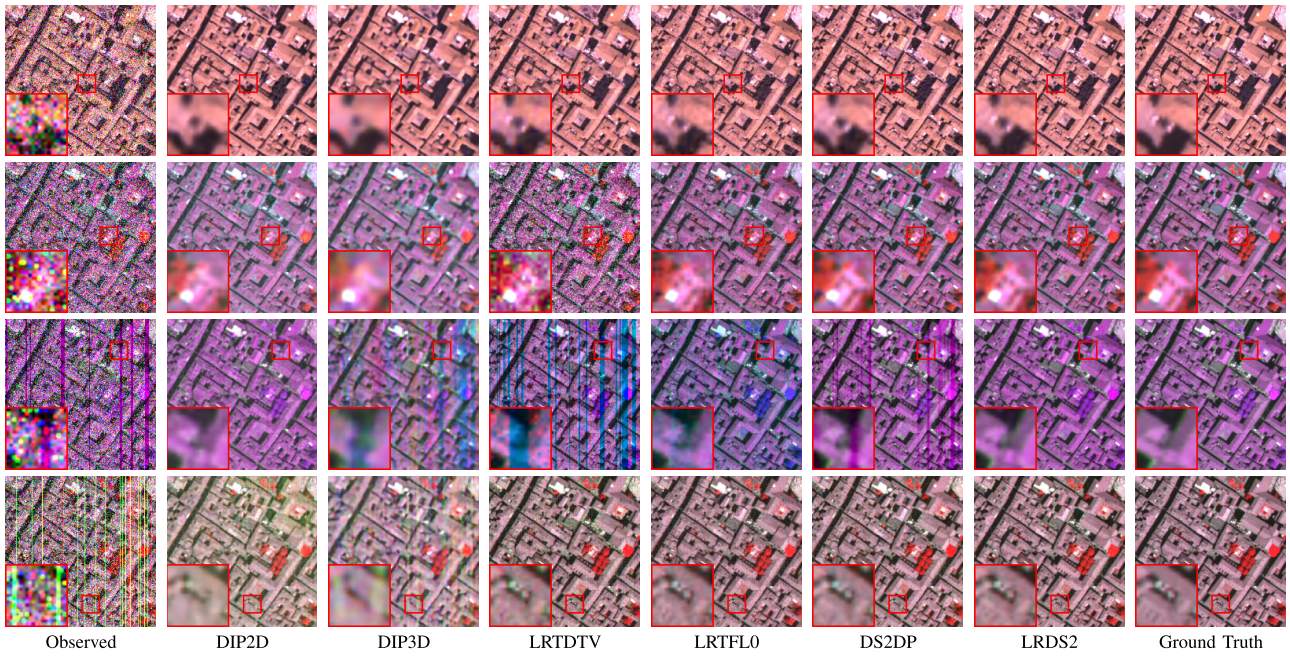


Fig. 4. False-color images of the denoising results by different methods on Pavia for Cases 1–4. From (top) to (bottom): the false-color images composed of bands $(32, 21, 14)$, $(62, 2, 21)$, $(54, 7, 61)$, and $(64, 27, 45)$, respectively.

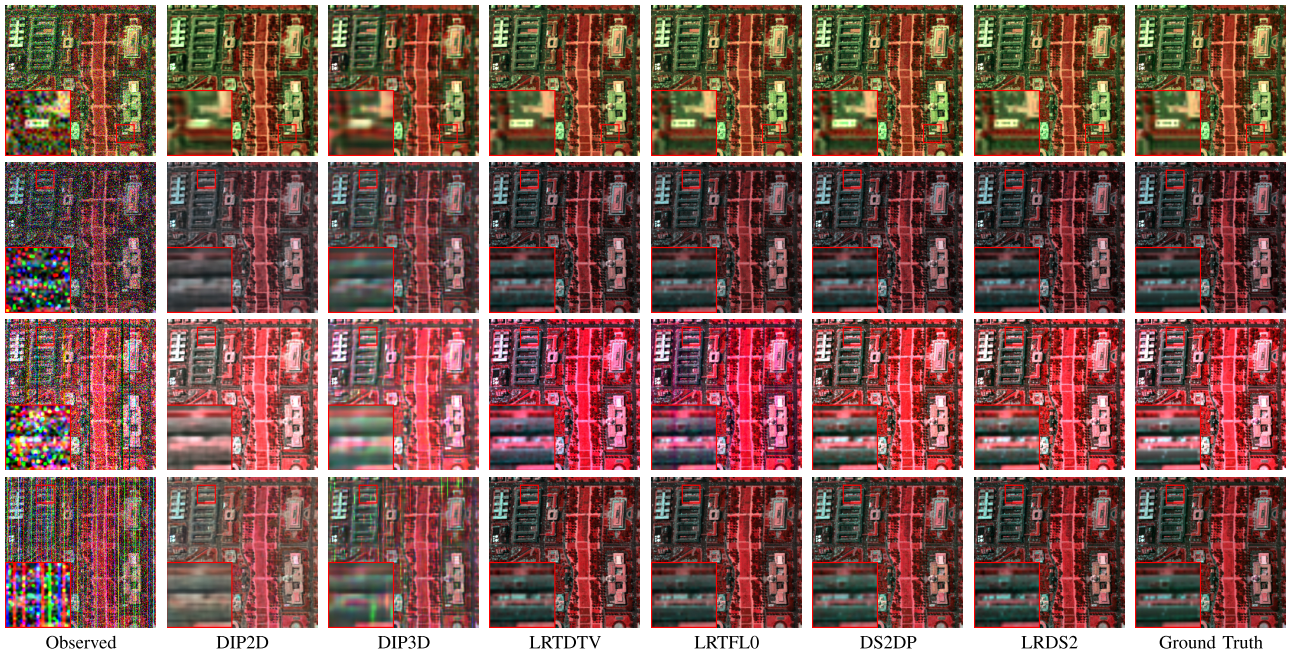


Fig. 5. False-color images of the denoising results by different methods on WDC for Cases 1–4. From (top) to (bottom): the false-color images composed of bands $(105, 11, 19)$, $(56, 1, 14)$, $(119, 11, 26)$, and $(125, 6, 21)$, respectively.

B. Real Data

In this section, we evaluate the performance of LRDS2 on two real HSI data. They are the Indian Pines[1] and GaoFen-5 (GF-5) datasets. Before our experiments, we clip the Indian Pines dataset from $145 \times 145 \times 220$ to $128 \times 128 \times 200$ and the GF-5 dataset from $2100 \times 2048 \times 180$ to $256 \times 256 \times 155$. The parameters β_1, β_2, μ, TR rank $[r, r, r]$, and M are set as 1, 100, 0.03, $[10, 10, 10]$, and 10 for Indian Pines and GF-5.

The false-color images of the denoising results by different methods on Indian Pines and GF-5 are shown in Fig. 8. For Indian Pines, we observe that the local details of the results by DIP2D, DIP3D, LRTDTV, LRTFL0, and DS2DP are oversmooth, especially in the enlarged region. However, the proposed LRDS2 can remove most noise and preserve better local details. We can also observe that there are still some stripes in the result of LRTDTV on GF-5. Some local details and textures of the results by DIP2D, DIP3D, LRTFL0,

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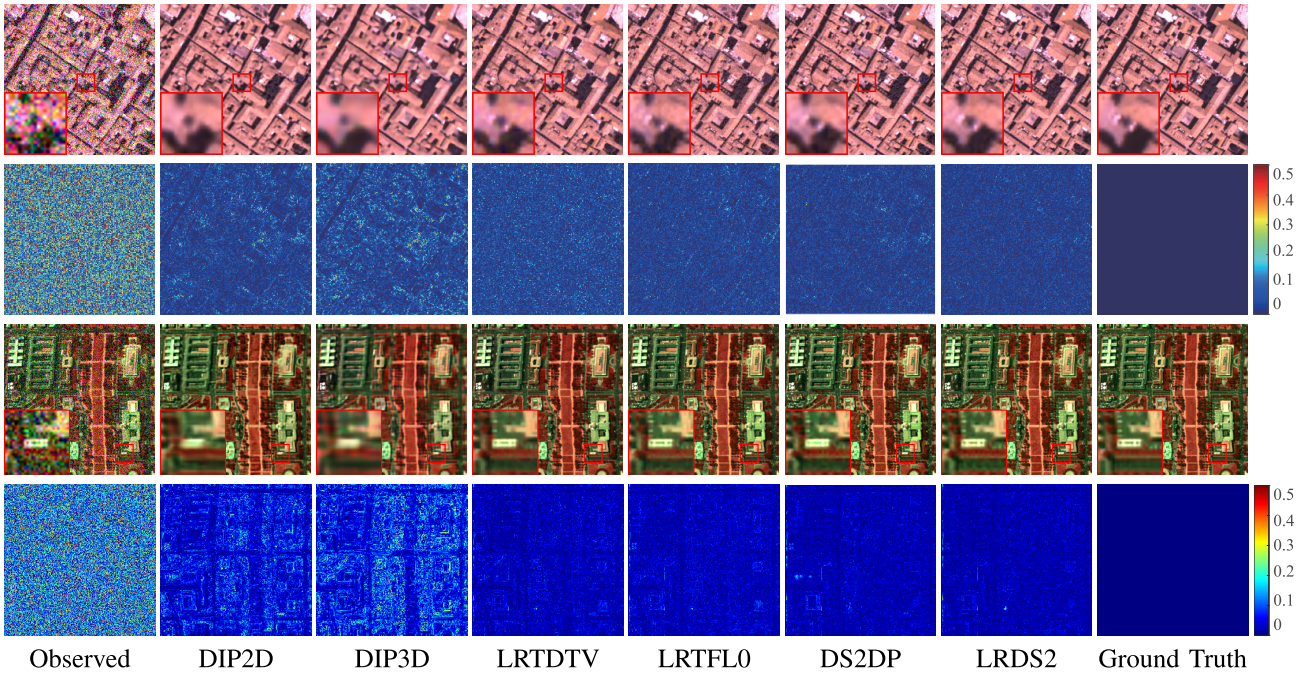


Fig. 6. Visual results by different methods on Pavia and WDC for Case 1. From (top) to (bottom): the false-color image composed of bands (132, 21, 14) on Pavia, the residual images of the results on Pavia on band 80, the false-color image composed of bands (105, 11, 19) on WDC, and the residual images of the results on WDC on band 100.

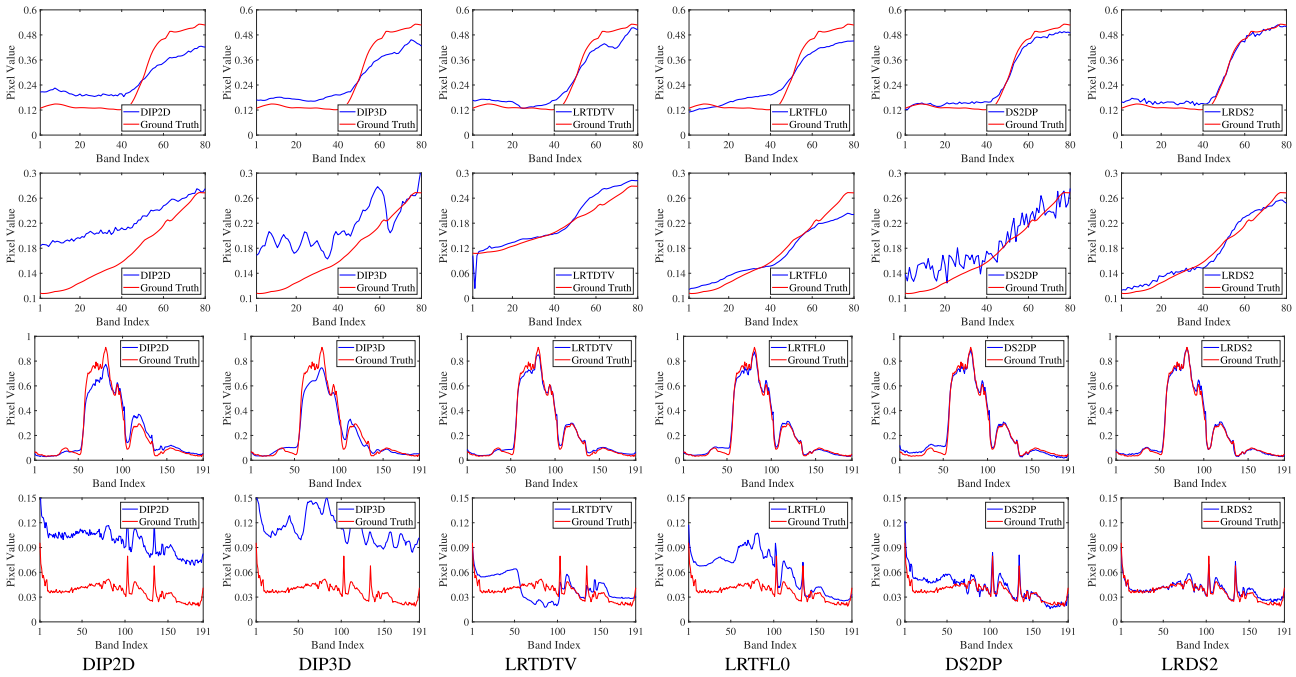


Fig. 7. Spectral vectors of the denoising results by different methods. From (top) to (bottom): the spectral vectors at the spatial location (2, 136) on Pavia for Case 1, the spatial location (5, 95) on Pavia for Case 2, the spatial location (80, 195) on WDC for Case 1, and the spatial location (115, 205) on WDC for Case 2, respectively.

and DS2DP are oversmooth and blurry, especially in the enlarged region. However, the result by LRDS2 preserves local details while removing most noise, which leads to better visual performance compared with other competing methods. These observations verify that the tensor low-rank prior and the deep spatial–spectral prior are complementary to each other, which leads to the superior performance of the proposed method in HSI denoising.

Fig. 9 shows the average of column vectors of the results by different methods on Indian Pines and GF-5. We observe that there are many rapid fluctuations in the average of column vectors of observed HSI. The average of column vectors of the

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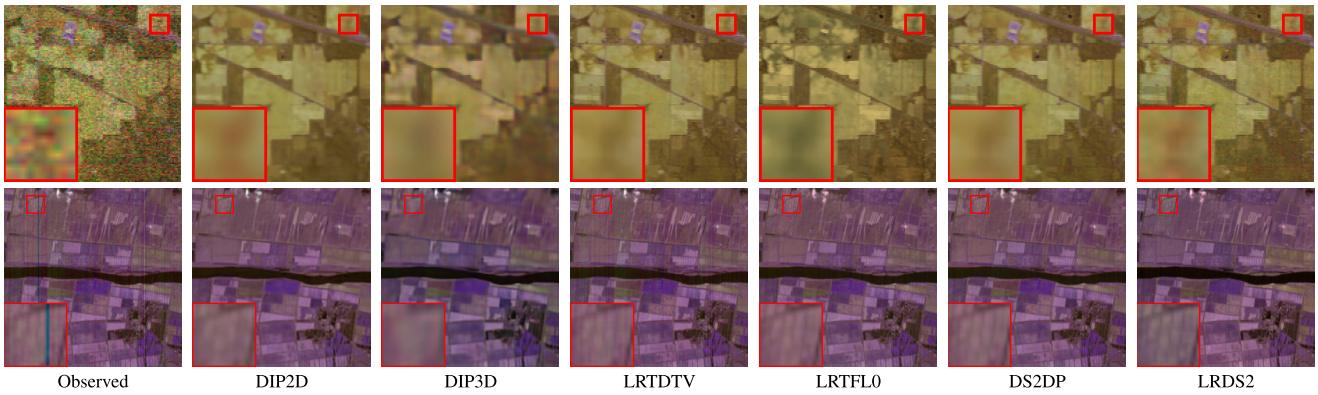


Fig. 8. False-color images of the denoising results by different methods on Indian Pines and GF-5. From (top) to (bottom): the false-color images composed of bands (104, 144, 3) on Indian Pines and bands (81, 95, 1) on GF-5.

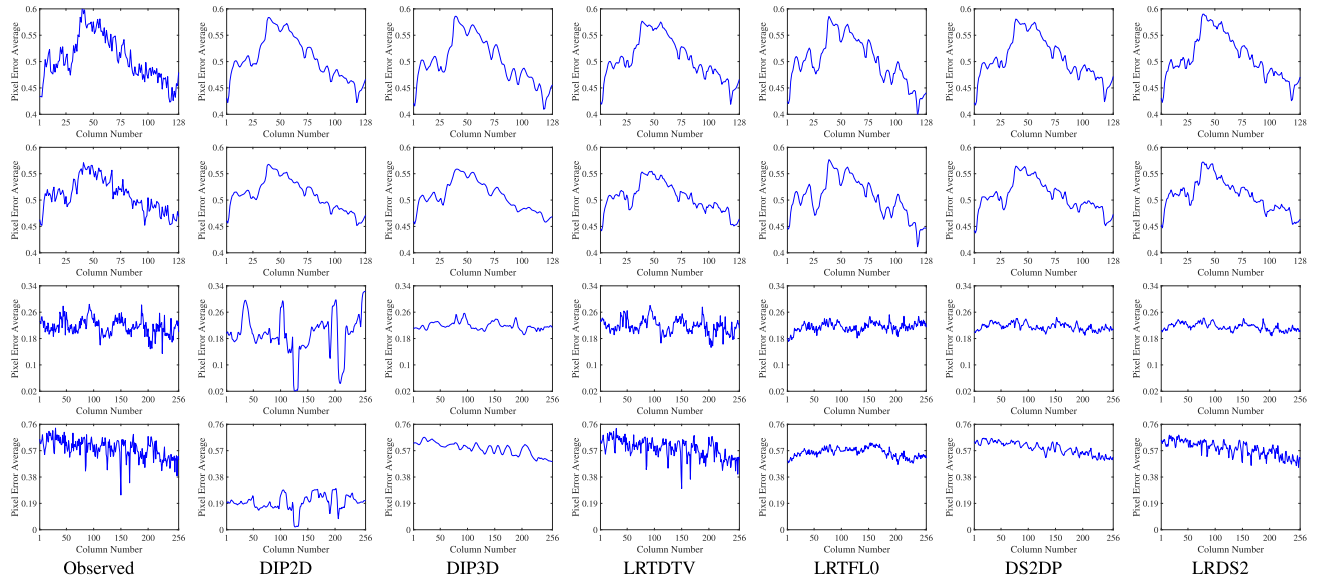


Fig. 9. Average of column vectors of the results by different methods. From (top) to (bottom): the average of column vectors of the results of the band 104 on Indian Pines, band 200 on Indian Pines, band 151 on GF-5, and band 155 on GF-5, respectively.

results by DIP2D, LRTDTV, LRTFL0, and DS2DP still has some rapid fluctuations. The averages of column vectors of the results by DIP3D are oversmooth. The results by LRDS2 are smooth, which implies better performance. These observations demonstrate that combing the tensor low-rank prior and the deep spatial–spectral prior can facilitate achieving better performance in HSI denoising.

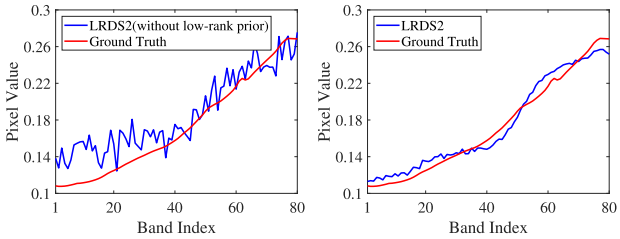


Fig. 10. Spectral vectors at spatial location (5, 95) of the results by LRDS2 without low-rank prior and LRDS2 on Pavia for Case 2.

V. Discussion

A. Ablation Study

In this part, we discuss the impact of the tensor low-rank prior, the deep spatial–spectral prior, and the sparse regularizer by some experiments on Pavia for Case 2.

1) Contribution of the Tensor Low-Rank Prior: To verify the contribution of the tensor low-rank prior, we employ LRDS2 without the tensor low-rank prior and the proposed LRDS2 for HSI denoising. Fig. 10 shows the spectral vectors of the results by different methods. From Fig. 10, we observe that the spectral vector of the result by LRDS2 is closer to the ground truth than that by LRDS2 without the tensor low-rank prior. The underlying reason is that the TR decomposition used to express the tensor low-rank prior can capture global spatial–spectral correlation well. The results demonstrate that the tensor low-rank prior can contribute to capturing the global structure of the underlying HSI.

2) Contribution of the Deep Spatial–Spectral Prior: To verify the contribution of the deep spatial–spectral prior,

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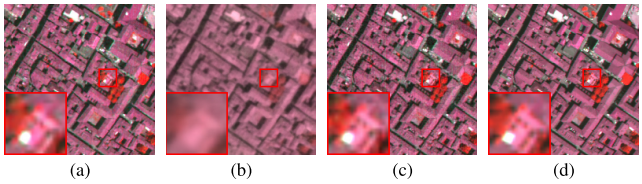


Fig. 11. False-color images composed of bands (62, 2, 21) of the results by different methods. (a)–(c) False-color images of the denoising results by LRDS2 without the deep spatial–spectral prior, LRDS2 without sparsity regularizer, and LRDS2. (d) Ground truth.

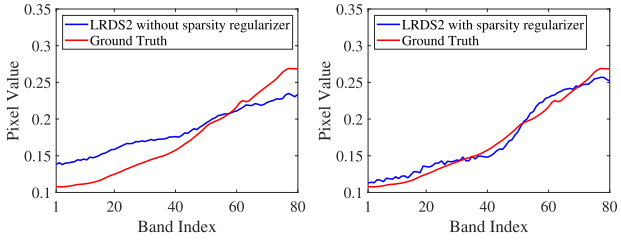


Fig. 12. Spectral vectors at spatial location (5, 95) of the results by LRDS2 without sparsity regularizer and LRDS2 on Pavia for Case 2.

we employ LRDS2 without the deep spatial–spectral prior and the proposed LRDS2 for HSI denoising. Fig. 11(a) and (c) shows the denoising results by different methods. From Fig. 11, we observe that LRDS2 preserves better local details than LRDS2 without the deep spatial–spectral prior. This is because DNNs that are used to characterize the deep spatial–spectral prior tend to capture the local details.

3) Contribution of the Sparsity Regularizer: To verify the contribution of the sparsity regularizer, we employ LRDS2 without sparsity regularizer and the proposed LRDS2 for HSI denoising. Fig. 11(b) and (c) shows the results by different methods. From Fig. 11, we observe that LRDS2 outperforms LRDS2 without sparsity regularizer on mixed noise removal. Fig. 12 shows the spectral vector of the results by LRDS2 and LRDS2 without sparsity regularizer. We observe that the spectral vector of the denoising result by LRDS2 is closer to the ground truth than LRDS2 without a sparsity regularizer. This result demonstrates that the sparsity regularizer in LRDS2 facilitates improving the performance in HSI mixed noise removal.

B. Sensitivity Analysis

In this section, we first discuss the sensitivity of all the parameters in LRDS2, which are β_1, β_2, μ, the numbers of endmembers M, and TR rank $[r, r, r]$, and then discuss the influence of the initialization of TR factors and the convergence of the proposed algorithm by conducting some experiments on Pavia for Case 1 and Case 2.

1) Influence of β_1: We conduct a series of experiments to discuss the sensitivity of β_1, which determines the weight of the TR decomposition-based regularizer term. Fig. 13 shows the PSNR values of the results by the proposed LRDS2 with different β_1's. From Fig. 13, we can observe that the PSNR values gain increment gradually with the increment of β_1, while the PSNR values decrease gradually when β_1 is larger

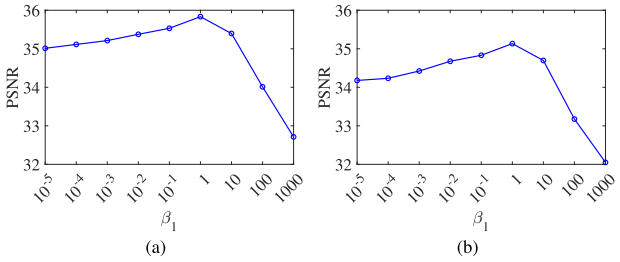


Fig. 13. PSNR values of the results by LRDS2 with different β_1's on Pavia for (a) Case 1 and (b) Case 2.

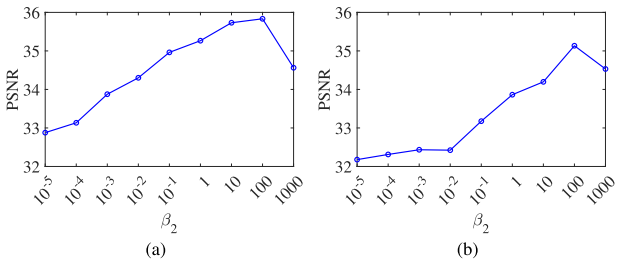


Fig. 14. PSNR values of the results by LRDS2 with different β_2's on Pavia for (a) Case 1 and (b) Case 2.

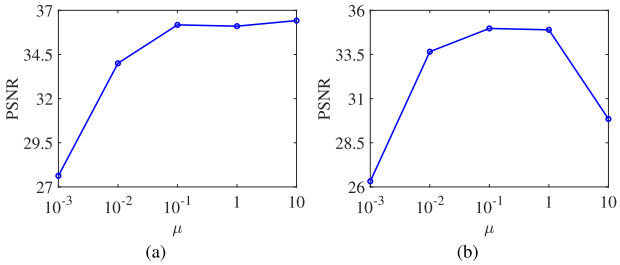


Fig. 15. PSNR values of the results by LRDS2 with different μ's on Pavia for (a) Case 1 and (b) Case 2.

than 1. Moreover, from Fig. 13(a) and (b), we can observe that the PNSR values have the same change tendency on different noise cases, which reveals the robustness of β_1 on different cases. Thus, the parameter β_1 is empirically set to be 1 in our method for Pavia.

2) Influence of β_2: We discuss the sensitivity of β_2, which determines the weight of the DNNs-based regularizer term. Fig. 14 shows the PSNR values of the results by the proposed method with different β_2's. From Fig. 14, we can observe that the PSNR values increase with the increment of β_2 when β_2 is smaller than 100, but, when β_2 is larger than 100, the PSNR values decrease gradually. Moreover, Fig. 14(a) and (b) shows that the change tendencies of PSNR values on different noise cases are similar, which implies that β_2 is robust to different cases. Hence, the parameter β_2 is empirically set to be 100 in our method for Pavia.

3) Influence of μ: Since μ is the key parameter used to restrain the sparsity of sparse noise, we conduct experiments by LRDS2 with different μ's to discuss its sensitivity. Fig. 15 shows the PSNR values of the results. From Fig. 15, we observe that the proposed method is sensitive to μ. Especially, the proposed method achieves the highest PSNR

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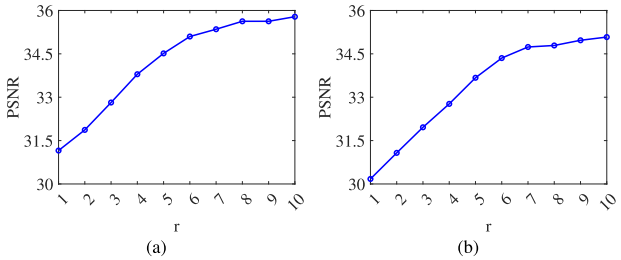


Fig. 16. PSNR values of the results by LRDS2 with different r's on Pavia for (a) Case 1 and (b) Case 2.

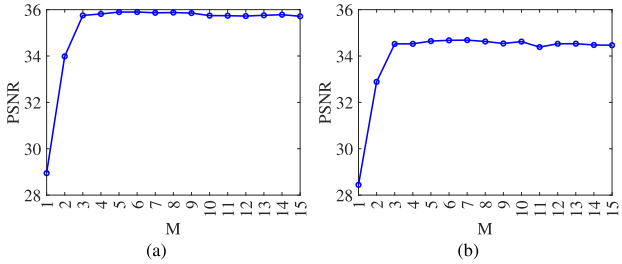


Fig. 17. PSNR values of the results by LRDS2 with different M's on Pavia for (a) Case 1 and (b) Case 2.

TABLE III

PSNR, SSIM, AND SAM VALUES OBTAINED BY LRDS2 WITH DIFFERENT INITIALIZATIONS OF TR FACTORS ON PAVIA FOR CASE 1. THE **BEST** VALUES ARE HIGHLIGHTED IN BOLD

Method	Initialization	PSNR	SSIM	SAM
LRDS2	Uniform	35.732	0.953	3.337
	Gaussian	**35.843**	**0.957**	**3.152**
	Constant	35.519	0.949	3.564

value on Pavia for Case 1 when $\mu = 10$ and the highest PSNR value on Pavia for Case 2 when $\mu = 0.1$.

4) Influence of r: Because r is the key parameter that determines the TR rank in the proposed method, we analyze its sensitivity through a series of experiments. Fig. 16 shows PSNR values for the results of the proposed method with different r's. We observe that there is a significant increase in the PSNR values with the increase in r. When r reaches 7, the PSNR values remain stable. In addition, the proposed method has similar change tendencies in different cases. Thus, the TR rank is empirically set to be $[7, 7, 7]$ in our method for Pavia.

5) Influence of M: We discuss the sensitivity of M, which is the number of the endmembers of the underlying HSI. Fig. 17 shows the PSNR values of the results by the proposed method with different M's. We observe that the proposed method has stable and superior performance when M is larger than 4. Under the guidance of this observation, the parameter M is empirically set to be 4 in our method for Pavia.

6) Influence of the Initialization of TR Factors: We discuss the influence of the initialization of TR factors by conducting some experiments. We initialize TR factors by the uniform distribution [e.g., $U(-1, 1)$], Gaussian distribution [e.g., $\mathcal{N}(0, 1)$], and constant value (e.g., 1). Table III lists the PSNR, SSIM, and SAM values obtained by the proposed method with different initializations of TR factors. From Table III, we can observe that the proposed method is robust with different initialization of TR factors. Guided by this observation, we utilize the Gaussian distribution to initialize the TR factors in our method.

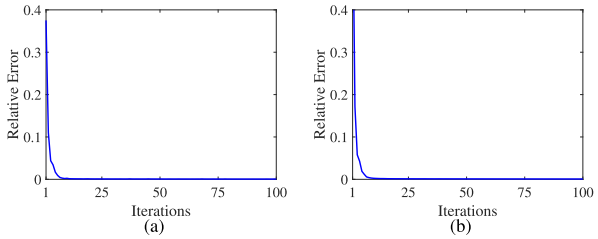


Fig. 18. Relative error curve of the alternating minimization algorithm versus the number of iterations on Pavia for (a) Case 1 and (b) Case 2.

7) Convergence Analysis: Since the proposed model is highly nonconvex and nonlinear, it is difficult to establish a rigorous convergence for the proposed algorithm under the current optimization framework. Instead, we display the curve of the relative errors of the alternating minimization algorithm in Fig. 18. From Fig. 18, we can observe that the relative errors quickly decrease to zero. This observation empirically verifies the numerical convergence of the proposed algorithm.

VI. CONCLUSION

In this article, we proposed an unsupervised HSI denoising method that integrates the tensor low-rank prior and the deep spatial–spectral prior to benefit each other. The deep spatial–spectral prior is expressed by DNNs, which can capture the local details of the underlying HSI. The tensor low-rank prior is characterized by TR decomposition, which can capture the global spatial–spectral correlation of the underlying HSI. To solve the proposed model, we developed an efficient alternating minimization algorithm. Compared with some state-of-the-art methods, extensive experiments demonstrated the superiority of LRDS2 for HSI denoising.

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